

Mariam Abdelfattah

Junior AI Engineer

+20 (109) 553-4805 | mariamalaa76200@gmail.com | [LinkedIn - mariamalaa76](#) | [GitHub - mariamalaa76](#)

PROFESSIONAL SUMMARY

Junior AI Engineer specializing in **Generative AI, LLMs, RAG, and intelligent agents**. Experienced in building end-to-end AI applications and deploying solutions using **Python, FastAPI, and AWS**. Strong foundation in machine learning and computer vision, backed by an **IEEE-published research paper on 3D Arabic Sign Language Recognition**.

EDUCATION

- Master's degree in information systems | Ain Shams University | Sept 2026 – Present
- B.Sc. in Computer Science | El Shorouk Academy | Oct 2022 – July 2026

TECHNICAL SKILLS

AI & Machine Learning: Generative AI, LLMs, RAG, AI Agents, Machine Learning, Deep Learning, Computer Vision

AI Engineering: Retrieval Pipelines, Prompt Engineering, Embeddings, Model Evaluation, API Integration, AI App Development

Data & Analytics: Data Preprocessing, EDA, Feature Engineering, SQL, ETL/ELT, Data Pipelines

Engineering & Deployment: REST APIs, FastAPI, Docker, CI/CD, AWS, MLflow

PROFESSIONAL EXPERIENCE

AI Agents & LLMs Intern (30 hrs, Onsite) | Orange Digital Center (Feb 2026 – Feb 2026)

- Developed **RAG-based AI agent workflows** connected to custom knowledge bases and external APIs.
- Built and deployed interactive AI applications, integrating **email and spreadsheet automation** to streamline workflows.

Machine Learning for Data Analysis Intern (120 hrs, Onsite) | NTI (Oct 2025 – Dec 2025)

- Built ML pipelines encompassing data cleaning, exploratory data analysis (EDA), and feature engineering.
- Trained and optimized neural networks, ensemble methods, and SVMs using TensorFlow and scikit-learn.

Coding Instructor (Hybrid – Part Time) | iSchool — DEMI & DECI National Projects (May 2024 – Aug 2026)

- Mentored **500+ students** in Python, data science, and ML by translating complex concepts into hands-on projects.

PROJECTS

Clinical RAG Assistant for Diabetes Guidelines | AI Hackathon

- Built an end-to-end **RAG-based clinical assistant** using trusted diabetes clinical guidelines to provide grounded answers to medical questions.
- Implemented **document processing, chunking, embeddings, Chroma retrieval, Top K evaluation, retrieval thresholding, and citation tracking** with page, section, and chunk-level metadata.
- Integrated the Google Gemini API for grounded answer generation and built a **FastAPI backend with a React UI**, including citation validation, insufficient-evidence handling, and patient-safety rules.

Tech Stack: Python, Gemini API, RAG, Chroma, FastAPI, React, Embeddings

EduPal AI — Intelligent Study Assistant | Generative AI & AWS Cloud

- Built an AI study assistant that processes PDF documents and generates adaptive quizzes using **AWS Bedrock**.
- Developed a PDF text extraction and context injection pipeline for **source-grounded LLM responses**.
- Deployed the application using **Streamlit and AWS**, with a serverless backend using **Lambda and API Gateway**.

Tech Stack: Python, AWS Bedrock, Lambda, API Gateway, Streamlit, PyPDF2, EC2

Telecom Customer Churn Prediction | Machine Learning & MLOps

- Built an end-to-end **ML pipeline** for data preprocessing, feature engineering, and training on 1M customer records.
- Handled severe class imbalance (9:1) using **XGBoost (scale_pos_weight) / SMOTE** with stratified CV, achieving **92% recall** and **0.88 ROC-AUC**.
- Added **MLflow** for experiment tracking and deployed the model through a **FastAPI** service using **Docker** and **GitHub Actions**.

Tech Stack: Python, XGBoost, Scikit-learn, FastAPI, MLflow, Docker, Joblib, Pandas, NumPy

PUBLICATIONS & RESEARCH

3D Alphanumeric Recognition in Arabic Sign Language | IEEE — ICAISET 2026 (April 2026)

- Developed the **computer vision pipeline** for a peer-reviewed IEEE publication, using **YOLOv8, WiLoR, and MANO** for 3D hand-pose feature extraction and classification, achieving **95.8% test accuracy** across 42 classes.

TECHNICAL TRAINING

- **Advanced Agentic AI (Sprints):** Completed hands-on training in LangGraph covering agent workflows, memory, tools, reasoning, RAG applications, HITL, and agent protocols.
- **AI & ML BootCamp (Sprints):** Completed 110 hours of hands-on project work building deep learning models, RAG pipelines, and agentic workflows.
- **Generative AI Trainee (NVIDIA DLI & ITI):** Built LLM-powered applications and RAG pipelines following NVIDIA DLI curriculum, with a focus on production deployment.
- **Data Engineering Training (ITI):** 120 hours covering Python, SQL, Linux, data preprocessing, ETL/ELT pipelines, and data warehousing.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Head of Machine Learning | MSP Tech Club ASU | Oct 2023 – Sept 2024

- Led a 25-member technical team, taught ML sessions, and organized a mid-year technical workshop.

HR Team Lead | MSP Tech Club ASU | Oct 2021 – Sept 2023

- Conducted 100+ candidate interviews and implemented structured KPIs to improve team performance.

Registration & Data Coordinator | NASA Space Apps Cairo Hackathon | May 2024 – Oct 2024

- Managed data operations and check-in logistics for over 500 hackathon participants.

Registration Team | ASU Career Center | Mar 2022 – Feb 2024

- Coordinated large-scale registration and logistics for employment fairs serving 10,000+ attendees.